$\boxed{\text{LECTURE 17}}$

Wednesday
1/4/26

Poison Distribution:

① Limiting approximation ~~of~~ of Binomial Distribution:
Case of Binomial distribution when probability of success (p) is very small (0.05) and no. of trails (n) is very large (>=30) approches to Infinity.

→ PDF: $\underbrace{f(x, \mu)}_{\text{parameters}} = \frac{\mu^x \cdot e^{-\mu}}{x!}$ $x = 0, 1, 2 \dots \infty$

Mean = Variance = μ
SD = $\sqrt{\mu}$
$\mu = n p$ (mean of Binomial distri.)

Q: If X is a piosson random variable with parameter $\mu = 2$; find prob. when $x = 0, 1, 2, 3$ or more.

$$p(x, 2) = \frac{e^{-2} \cdot (2)^x}{x!} \quad (x = 0, 1, 2 \dots)$$

$p(0,2) = e^{-2} = \boxed{0.135335}$
$p(1,2) = (e^{-2} \cdot 2)/1! = \boxed{0.27067}$
$p(2,2) = (e^{-2} \cdot 2^2)/2! = \boxed{0.27063}$
$p(X \geq 3) = 1 - P(X < 3)$ compliment rule since greater than 3.
$= 1 - [P(0,2) + p(1,2) + p(2,2)]$
$= 1 - [0.135335 + 0.27067 + 0.27063]$
$= \boxed{0.353325}$

Q: 200 have made reservation of flight. If prob. that a passenger who has reservation won't show up is 0.01, what is prob. that exactly 3 will not show up.

- $n = 200$, $p = 0.01$ (didn't show) ∴ since 'n' is very large as compared to 'p'; we'll use piosson distri
- $\mu = 200 \times 0.01 = \boxed{2}$
- $P(X=3) = P(3, 2) = \dfrac{(2)^3 \cdot e^{-2}}{3!}$

$$= \boxed{0.1804}$$

Q: Out of 100 items; 1% is defective. Let X be no. of defective found in items of 100. Construct a table of Prob. for $x = 0, 1, 2, 3, 4$ by Binomial & piosson distribution

$n = 100$, $p = 0.01$, $q = 0.99$

→ Binomial:

$$P(X = r) = {}^{100}C_r \cdot (0.01)^r \cdot (0.99)^{100-r} \quad (r = 0, 1, 2, 3, 4)$$

→ Poisson: $\mu = 100 \times 0.01 = \boxed{1}$

$$P(X = x) = P(x, 1) = \frac{1^x \cdot e^{-1}}{x!} \quad x = 0, 1, 2, 3, 4.$$

x	Binomial	Poisson
0	0.3360	0.3679
1	0.3697	0.3679
2	0.1849	0.1839
3	0.0610	0.0613
4	0.0149	0.0153

Since results of Both are very close so it's a good approx.
Binomial always gives exact
Piosson gives approx.

② Poisson Process : Event's occurence w.r.t Time/space/Length.

PDF: $p(x, \lambda t) = \dfrac{e^{-\lambda t} \cdot (\lambda t)^x}{x!}$

x → no. of occurences in t unit of time

t → no. of units of time

λ → avg. no. of occurence per unit time

Q: Telephone calls are being placed through a certain exchange at random times of the average of the four per minute. Determine prob. that in a 15 second interval there are 3 or more calls. Since we've average event occurance (λ) given in ques. ∴ it's a poisson process question.

$\lambda = 4$ calls per minute (60 seconds) $\longrightarrow \lambda t = 4$

$\lambda = 4$ calls per 15 seconds $\longrightarrow \lambda t = 4 \cdot \frac{15}{60} = \boxed{1}$

$p(x, \lambda t) \rightarrow p(X \geq 3) = 1 - p(x, \lambda t)$ $\quad x = 0, 1, 2.$

$= 1 - \left[\sum_{x=0}^{2} p(x, \lambda t)\right]$

$= 1 - \sum_{x=0}^{2} \dfrac{e^{-1}(1)^x}{x!} = \boxed{0.08025}$

General Formula: $t = $ new/old

new → jo pucha gaya ho

old → jo ques. main me average kay sath given ho.

new = 15 seconds

old = 60 seconds (it was 1 minute but we changed it in seconds aswell)

$= 15/60 = \boxed{1/4}$ $\quad$ ∴ $\lambda t = 4 \cdot (1/4) = \boxed{1}$.

Q: Average of one in 150 (old) square feet .... find prob. of at most one flaw in 225 (new) square feet. Finding 't'

$t = $ new / old.

$t = 225/150 = \boxed{1.5}$